MVA Hackathon 2026 ??? Track 2 Report

Maaz Asher Khurram ?? mk7790@nyu.edu

Repurposing candidates for a child with BUB1B-associated mosaic variegated aneuploidy

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Repurposing candidates for a child with BUB1B-associated mosaic variegated aneuploidy

Rare Disease, Real Kid: The MVA Hackathon 2026 — Track 2 (Drug Repurposing)
Maaz Asher Khurram — mk7790@nyu.edu
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> **These are hypotheses for follow-up, not evidence that any medicine works.** Nothing here is clinical advice, and no proposal below has been tested in mosaic variegated aneuploidy or in this child.

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1. Summary

We carry **two therapeutic theses**. They are ranked, and the ranking changed late in this work when we asked which one is actually aimed at this child's lesion.

Lead — Thesis B: raise the amount of BubR1 the child still makes. Allele 2 (`p.Asn1002Lys`) is a **folding-class** variant: the protein is made, misfolds, and is cleared by chaperone-dependent quality control. That is a druggable axis with approved agents, and it is the only route where mechanism and paediatric safety currently coincide. The candidate is **arimoclomol**, an HSF1 co-inducer FDA-approved as Miplyffa on 20 Sep 2024 for Niemann–Pick type C in children aged two and over.

This report's first draft named sodium phenylbutyrate (4-PBA) instead, and the reversal is the clearest single illustration of how it was audited. 4-PBA was chosen partly *because* it is an HDAC inhibitor: we counted that as a third mechanism pointing the same way. It points the other way. HDAC inhibition induces premature sister chromatid separation and strips BubR1 from kinetochores (Magnaghi-Jaulin 2007; Dowling 2005), and MVA1's other name is **premature chromatid separation syndrome**. Either the activity is live at achievable exposure and the drug partially phenocopies the disease, or it is not live and the third leg was never there. There is no reading in which it counts in favour. 4-PBA remains the better-characterised paediatric agent and is kept as a stated alternative with that hazard attached; arimoclomol carries the matched mechanism without it (`CORRECTIONS-2026-10-06.md`, C5 and C7).

Adjunct — Thesis A: blunt the senescence-associated secretory phenotype. Anakinra and canakinumab, reached by a six-gate screen of **5,202 molecules** in which three independent instruments converged on the IL-1 axis. This was our original lead. It is retained because the screen behind it is sound and the convergence was not engineered, but it treats a *downstream consequence*; Thesis B treats the lesion.

Two findings make Thesis B specific to this disease rather than borrowed from another protein. First, aneuploid cells **lose HSF1 and HSP90** and cannot mount a heat-shock response — so this child's aneuploidy depletes the very folding capacity that would buffer a destabilised BubR1, and a chaperone co-inducer restores a depleted system rather than augmenting a normal one (§3.3). Second, the field excludes chaperone candidates when the active site is affected; **N1002 sits 26.7 Å from the ATP site**, so the pocket is intact and the lesion is purely in the fold — the configuration this drug class is built for (§3.2).

And one thing is ruled out rather than merely untested. BubR1's pseudokinase domain has **no ligandable allosteric site that is not already a functional site** — a 757-fragment screen on a validated instrument found its one non-functional groove indistinguishable from bare protein (§3.4). That closes the *pharmacological* chaperone route, which binds the target directly. It leaves the lead intact, because neither candidate binds BubR1: both act in ***trans*** on folding capacity — 4-PBA as a chemical chaperone, arimoclomol through HSF1 — a distinction this report is careful to keep.

Why the order flipped. Thesis A's premise rests on two human datasets that disagree (§5). Thesis B's premise — that N1002K is a folding variant — is the one we could test, and did: within the folding class, stability prediction separates pathogenic from benign at **AUC 0.879–0.920** against a burial-matched null (permutation $p \leq 0.0052$ at every matching window), where pooling mechanism classes had destroyed the signal entirely (AUC 0.714, *below* burial alone). See `SYNTHESIS.md`.

Self-assessed verdicts, reported separately:

	thesis	verdict	the binding uncertainty
B	allele-2 stabilisation	**mechanistically grounded, magnitude probably insufficient; carries a tumour-promotion hazard**	dose anchor retracted (C6); BubR1 is an RMS survival dependency (C4)
A	IL-1 / SASP blockade	`interesting_under_supported`	two human datasets disagree by cell type
C	aneuploidy-selective killing (KIF18A)	**refuted on one of two original grounds**	SAC-routed killing holds; the "no euploid compartment" ground was retracted (C2)

A third arm, and it is the one the clinical record forces. `CHEMOSENSITIVITY-ARM.md`. Antimitotic chemotherapy kills by triggering **checkpoint-dependent mitotic arrest** — and this child's checkpoint is genetically crippled, which is the diagnosis. Vincristine is a backbone of rhabdomyosarcoma therapy, the tumour this child has already had. Germline *BUB1B* variants that **reduce BubR1 expression and stability** — precisely this lesion — drive taxane resistance via premature mitotic exit, and direct *BUB1B* knockdown shifts antimitotic IC50 roughly tenfold.

That reframes the task from *what drug can we add* to **which approved drugs will and will not work in this genotype** — and it gives the lead a second, far more measurable rationale: if low BubR1 blunts checkpoint-dependent chemotherapy, then raising it is potentially **chemosensitising**, testable as an adjunct rather than only as a chronic disease-modifier. The honest limits are set out in full in that document; the largest is that the direct drug evidence is taxane rather than vincristine, which is a cheap experiment away.

A fourth arm, tested and refuted — and the refutation is the most useful result in this section. `ANEUPLOIDY-SELECTIVE-ARM.md`. Rather than fix the checkpoint, kill the chromosomally unstable cells: **KIF18A** inhibitors do exactly that, are in clinical trials, and spare chromosomally stable cells. It fails here twice over. First, KIF18A inhibition kills *through* SAC activation — and the published rescue experiment in the osteosarcoma study is **BubR1 siRNA**, this child's lesion, which abolishes the killing. Our second original ground — that CIN-selectivity needs a euploid compartment to spare — is **retracted** (C2): the real axis is transformed vs untransformed (Janssen 2009: untransformed fibroblasts could not be sensitised), and MVA is mosaic, so a euploid fraction exists. What survives is narrower and still useful: a selectivity argument must name the contrast it actually relies on.

Together those two arms state one rule, derived from the genotype rather than from any single drug:

> **The direction of effect of SAC deficiency on cytotoxic therapy is not fixed. It is conditioned on the observation window, on achieved drug concentration, and on which of BubR1's two functions — checkpoint signalling or chromosome alignment — the lesion impairs.**

An earlier version of this section stated a one-directional rule: that any SAC-routed therapy would underperform here. **That is retracted** (`CORRECTIONS-2026-10-06.md`, C1). Three primary sources carry the opposite sign — Lee 2004 (robust arrest *is* resistance; BubR1 RNAi *reversed* paclitaxel resistance), Janssen 2009 (taxol sensitisation came specifically from reducing BubR1, a dual-function protein, and *not* from reducing Mad2), and Dowling 2005 (checkpoint inactivation potentiates microtubule-drug killing).

Cohen-Sharir 2021 (Nature 590:486–91) resolves the apparent contradiction in the cell population that matters here: aneuploid cells are *less* sensitive to **short-term** SAC inhibition but become *increasingly* sensitive **over time**, because they keep dividing and accumulate unstable, less-fit karyotypes. The sign flips with the window. Our original rule took the 72-hour sign and extrapolated it to clinical response, which is precisely the extrapolation that paper forbids.

The symmetric half, which we had never written down: any chronic agent that *degrades* SAC function is contraindicated in a SAC-deficient child. Applied honestly it convicts our own lead — 4-PBA is an HDAC inhibitor, and HDAC inhibition induces premature sister chromatid separation and strips BubR1 from kinetochores (C5). MVA1's other name is premature chromatid separation syndrome.

Every high-risk intervention ever reported in this syndrome supports the same conclusion, and that is stronger than the statistics we had been citing (C28b). `MVA-THERAPEUTIC-EVIDENCE-BASE.md` tabulates them, and they split by *what they aimed at* rather than by how aggressive they were. The ones that **removed or treated a consequence** worked: tamoxifen gave sustained complete remission of a *MAD2L1BP* granulosa cell tumour; surgery alone carried a *MAD1L1* patient through 12 neoplasias to no new pathology since 2014; a renal transplant at 20 under ciclosporin was tolerated in *BUB1B* MVA1. The ones that tried to **repair or replace the system** failed: growth hormone in *CEP57* was ineffective and regretted in print, G-CSF and romiplostim gave "no response", and antiepileptics did not control the seizures.

The sharpest case is HSCT in MVA1 — the same gene as this child. A TCRαβ/CD19-depleted matched-unrelated graft on treosulfan-based reduced-intensity conditioning **achieved its mechanistic goal**: it engrafted, was well tolerated, and **cleared the monosomy 7 clone**. Then graft rejection on day +47, pancytopenia, multi-organ sepsis, and **death on day +156**. The authors state plainly that this *"needs to be taken into consideration for HSCT in other MVA patients."* It is not a treatment that failed to work; it is a treatment that worked and whose own complications killed the patient. Any proposal in this syndrome has to be read against that.

Neither thesis outranks surveillance, and the clinical record strengthens that. This child's presenting event was **rhabdomyosarcoma** — the predicted tumour type, already realised (§2) — so what is owed is *survivor* surveillance for relapse and second primaries rather than prophylaxis. `STANDARD-OF-CARE.md` establishes that expected benefit from LFS-style tumour surveillance done well — 5-year survival 59.6% → 88.9%, in a tumour spectrum that clusters in the first three years of life — is almost certainly greater than from any repurposed drug we can currently propose. If one thing is acted on, it should be surveillance. Everything below is hypothesis generation for follow-up, ranked honestly, and it does not compete with that.

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2. Variant mechanism

Genotype. Compound heterozygous *BUB1B* (chr15q15.1), consistent with mosaic variegated aneuploidy syndrome 1 (MIM 257300, MONDO_0009759).

Allele	Coordinate (GRCh38)	HGVS	Effect	Evidence
1	chr15:40,209,701 T>G	`c.2210T>G`	**p.Leu737Ter** — truncating	rs759242053; ClinVar **pathogenic** for MVA1
2	chr15:40,220,612 T>G	`c.3006T>G`	**p.Asn1002Lys** — missense	this **nucleotide** change is not in ClinVar, but the **same protein change** is: VCV004600147.1 (`c.3006T>A`), **VUS**, 19 Sep 2025 (C16); AlphaMissense **0.9229, likely_pathogenic** — but see C43: AlphaMissense fails its direction check on this protein's measured panel, so this score is reported for completeness and is **not** treated as evidence

Both were confirmed by Ensembl VEP against transcript ENST00000287598, **and both are present in the proband's own VCF** — heterozygous and PASS, at high confidence (`proband/confirm_variants.py`). The derivation and the child's data agree. Allele 2 is absent from ClinVar, so before running VEP we independently derived it by reconstructing the reading frame from three annotated neighbours (R999Q at 40,220,602; N1004S at 40,220,617; D1005E at 40,220,621), which agree to the base pair. The reconstruction and VEP give the same answer.

Direction of effect: loss of function. Allele 1 truncates 314 residues from a 1,050-residue protein. Domain annotations on ENSP00000287598 place the BubR1 kinase domain at **750–1045** (CDD `cd14029` STKc_BubR1_vert) or **782–1042** (Gene3D 1.10.510.10), with a tighter kinase-like core at 791–984 (SuperFamily SSF56112). p.Leu737Ter therefore stops translation **immediately upstream of the kinase domain on every annotation, removing all of it** — not merely part.

Allele 2, p.Asn1002Lys, falls **inside** the domain on the BubR1-specific CDD model and on Gene3D, but **outside** the tighter SuperFamily core. The structure settles what the annotations leave ambiguous: in AlphaFold `AF-O60566-F1`, **residue 1002 has pLDDT**

91.1 — "very high" confidence, sitting in a well-ordered region (kinase domain mean 82.8). Residue 737 has pLDDT 73.6 and sits at the boundary of a disordered linker (residues 600–749 mean **44.8**). So p.Leu737Ter truncates at the end of a disordered segment immediately before the ordered catalytic module and removes all of it, and p.Asn1002Lys substitutes a residue in a confidently-folded part of that module — which is where a missense substitution is most likely to be structurally consequential. The Mad3/Bub1 homology region (Pfam PF08311, 57–179) is unaffected by either allele.

Transcript-level evidence (AlphaGenome). Both variants were scored with AlphaGenome over a 524 kb window centred on BUB1B, against a null built from 12 random SNVs inside the same gene:

scorer	null median	null max	L737*	N1002K
RNA_SEQ	0.0132	0.0202	0.0101	**0.0476**
SPLICE_SITES	0.0059	0.0078	**0.0244**	0.0117
SPLICE_JUNCTIONS	0.0318	0.0870	**0.1769**	0.0722

p.Leu737Ter scores **above the within-gene null maximum** on both splice scorers — roughly 2x on junctions and 3x on splice sites. That is consistent with a transcript-level consequence, most plausibly **nonsense-mediated decay**, which matters here specifically: NMD is what makes allele 1 a true null rather than a stable truncated protein, and the null-plus-hypomorph argument below depends on that distinction. p.Asn1002Lys's strongest signal is expression, also above the null maximum.

Two limits travel with this. Absolute effect sizes are small (0.01–0.18) and the null is only 12 variants, so this is suggestive rather than established. And AlphaGenome scores regulatory and splicing effects, **not** protein-level pathogenicity — a low score is never evidence that a nonsense variant is benign.

Evolutionary constraint (GPN-Star). A third predictor, orthogonal to the other two: GPN-Star (Ye et al., *Nature* 2026, DOI) scores constraint learned from whole-genome alignments with an explicit species tree. Both variants were looked up in the authors' precomputed genome-wide log-likelihood ratios and ranked against **every other possible SNV in BUB1B** (n = 180,342), on all three published human timescales — the paper reports task-dependent advantages of recent versus deep evolution, so no single timescale is privileged:

	primate 243-way	mammal 447-way	vertebrate 100-way
L737*	−6.35 (1.3rd %ile)	−6.73 (1.5th %ile)	−6.87 (1.3rd %ile)
N1002K	−5.68 (1.9th %ile)	**−7.78 (0.9th %ile)**	−7.42 (1.0th %ile)
null median	*−0.27*	*−0.78*	*−0.20*

Both variants fall in the most-constrained 1–2% of all possible BUB1B SNVs on every timescale, roughly ten to thirty times more negative than a typical variant in the gene. N1002K is markedly more constrained on the deeper mammal and vertebrate alignments than on primates alone, which is what deep conservation of a residue looks like — consistent with its position in the confidently-folded kinase core above.

So three predictors trained on three different signals — protein-level (AlphaMissense), transcript-level (AlphaGenome) and evolutionary (GPN-Star) — agree on both alleles. None of the three was selected after seeing the others.

That agreement is worth less than this report originally claimed, and the reason is measured. AlphaMissense was tested on this protein's own measured panel — five variants with 5–10× reduced BubR1 by western, two measured normal — and it ranks only **one of the five** above both normals. `Q921H`, measured normal, outscores four of the five measured-destabilising variants; `R814H`, measured at 5–10× reduced, scores 0.22 (`missense_arm.py`, CORRECTIONS C43). A predictor that points the wrong way four times in five on this protein's measured variants does not get a vote on allele 2, so **AlphaMissense's 0.9229 for N1002K is withdrawn as evidence** and the agreement above rests on two predictors, not three.

The verdict hinges on `Q921H`: drop it and all five destabilising variants clear the remaining normal, turning 1/5 into 5/5. There is a real case for dropping it — it sat on the same allele as `L844F`, the authors attributed causality to `L844F`, and it was never tested as the only variant on an allele. It is kept anyway, for the same reason §3.5 keeps the weaker posterior: discarding the one inconvenient data point is not how a number is obtained.

The null-plus-hypomorph architecture is what makes MVA1 survivable — biallelic complete *BUB1B* loss is embryonic lethal — so a viable patient essentially must carry one truncating and one partial-function allele.

Pathway disrupted. BUB1B (BubR1) is a core component of the **spindle assembly checkpoint** (SAC), which delays anaphase until every kinetochore is correctly attached. Loss of SAC function permits anaphase with unattached or mis-attached chromosomes, producing the constitutional mosaic aneuploidy that names the disease.

Downstream consequence, and why it is genotype-specific. Not all MVA carries tumour risk. Yost et al., *Nature Genetics* 2017 (DOI) state that "individuals with biallelic TRIP13 or BUB1B mutations have a high risk of embryonal tumors, and here we show that their cells display severe SAC impairment. MVA due to biallelic CEP57 mutations, or of unknown cause, is not associated with embryonal tumors and cells from these individuals show minimal SAC deficiency." **Risk tracks the severity of checkpoint impairment.** This child is in the high-risk group by genotype, not by inference. The associated tumour spectrum is embryonal broadly — Wilms tumour, **rhabdomyosarcoma**, leukaemia, Sertoli–Leydig tumours — and the association is constitutional rather than somatic: Hanks et al., *Cancer Letters* 2005 (DOI) screened 30 Wilms tumours, 10 ALLs and 20 rhabdomyosarcomas and found somatic *BUB1B* mutations "unlikely to be common in sporadic childhood cancers."

The prediction is realised, not prospective — and that is the strongest genotype–phenotype evidence in this report. The clinical record (`PHENOTYPE-GROUNDING.md`, HPO terms) shows the proband's presenting event was **rhabdomyosarcoma (HP:0002859)** — the malignancy that triggered urgent investigation. That tumour type sits inside the embryonal spectrum Yost assigns specifically to biallelic *BUB1B* with severe SAC impairment. So the chain *genotype → severe SAC impairment → embryonal tumour* is not an inference this report makes about a child at risk; **it is an observation about a child in whom it already happened.** The record's other features corroborate the rest of the model: intrauterine growth restriction (HP:0001518), short stature (HP:0004322), failure to thrive (HP:0001508) and skeletal muscle atrophy (HP:0003202) are the growth-restriction picture BubR1 hypomorph mice show, which is the dose–phenotype relationship §3.1 reasons from. And **parental recurrent spontaneous abortion (HP:0200067)** is independent support for a heritable biallelic mechanism — each parent carrying one allele — which is what the unphased *trans* inference needs and what a *cis* arrangement would not produce.

Three things this changes downstream, which earlier drafts did not carry. Cancer risk is **realised**, so surveillance shifts from prophylaxis to *survivor* surveillance for relapse and second primaries. A child treated for rhabdomyosarcoma has had **cytotoxic therapy**, which bears directly on the IL-1 arm's infection risk (§4.2) and on marrow reserve. And **nephrocalcinosis from birth (HP:0000121)** is a renal finding that any chronic agent in a small child has to answer to — including 4-PBA's sodium load (§3.5).

The constraint this places on any repurposing proposal, and the escape from it. A missing checkpoint cannot be pharmacologically restored — there is no agonist for a truncated kinase. That reasoning drove the original search entirely toward *downstream consequences*, and we pursued three: aneuploidy-induced senescence and its secretory phenotype; cancer predisposition; and a ciliopathy axis (Miyamoto & Matsuura, *Oncotarget* 2015, DOI). That became §4.

But the premise is only half true, and the half it misses is the lead. It holds for allele 1, which is a true null. It does **not** hold for allele 2: `p.Asn1002Lys` is a missense variant in a confidently-folded domain, so the protein is *made*. If it is destabilised rather than catalytically dead, the target is not a missing function to be agonised but a **protein abundance to be raised** — and that is upstream of every downstream consequence at once. §3 tests that premise and follows it.

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3. The lead — raise the protein the child still has

A missing checkpoint cannot be restored pharmacologically: there is no agonist for a truncated kinase. But this child is not checkpoint-null. Allele 2 makes protein. The question Thesis B asks is whether we can make it make **more**.

3.1 How much more, and can either allele supply it?

Taking one wild-type allele as 50 units of a 100-unit diploid normal, the estimated baseline here is **~17 units** (`estimate_bubr1_dosage.py`). Two targets bracket the ask: **~35 units** is the milder published comparator where the phenotype is attenuated (**2×), and 50 units** is carrier-equivalent, where MVA does not appear at all (**~3×**). Full derivation in `WHICH-ALLELE-TO-TARGET.md`.

route	ceiling	feasible in this child?
Allele 1 — **readthrough drug**	**+0.5 units** against a needed +18	**No — dead on arithmetic**
Allele 1 — **NMD inhibition**	**+40 units**	best ceiling; no safe drug exists
Allele 2 — **stabilisation**	needs ~2×	**the only open route**

Readthrough is not merely unproven here — it is quantitatively insufficient, by ~35-fold. Across ~5,800 pathogenic PTCs measured head-to-head, median readthrough ranges from 0.08% (gentamicin) to 1.32% (SJ6986); the *top decile* reaches 4.28% at best. This child's PTC is a **poor substrate on three of the three known context rules** — the +1 nucleotide is A, the −1 is G, and the upstream codon is GAG, all unfavourable. Only the UGA stop identity favours readthrough, and we had previously over-read that single feature. And readthrough acts only on NMD survivors: the PTC sits **747 nt** upstream of the last exon–exon junction, exceeding the 50–55 nt rule thirteenfold, flanked by two

truncations empirically confirmed transcript-null in patients (731X, 753X). **Crossed off**, and with it DAP, ELX-02, and ataluren — whose clinical failure now looks like what the biochemistry predicts.

NMD inhibition has the only large ceiling and two blockers. No NMD inhibitor is approved anywhere for NMD inhibition. **Amlexanox is better credentialled than this report previously said, and the earlier text — "a topical/short-course agent, not chronic systemic dosing in an infant" — is withdrawn (23 Oct 2026).** It has decades of approved use, an oral indication in Japan, and has completed a randomised, double-blind, placebo-controlled **oral** trial in 42 adults (Oral et al. 2017, *Cell Metab*, PMID 28683283); it rescues **UGA** PTCs functionally in patient-derived neuronal cells at 100 µM (PMID 37513945). The real blockers are different and are set out in `ALLELE1-APPROVED-DRUGS.md`: the measured free Cmax (0.11–0.29 µM) falls short of every reported effective concentration — by ~90–900× against the 25–100 µM TBK1/TFEB separation, while a claimed 0.4 µM activity that would have narrowed the gap to ~1.4–3.6× could not be traced to any indexed publication, so it is recorded as an open uncertainty rather than a correction (**C51**), amlexanox separately inhibits the innate-immune kinases TBK1/IKKε, and NMD inhibition remains global. And the mouse data place the risk exactly where this child is vulnerable — **neurological tissues were the most responsive to NMD change**, in a child of this age on chronic systemic dosing. **Correction:** earlier drafts justified this by saying the child's phenotype is dominated by microcephaly and developmental delay. The clinical record lists neither — that was imported from the general MVA literature and the hypomorph mouse (`PHENOTYPE-GROUNDING.md`). The neurotoxicity concern stands on the mouse tissue data and on chronic dosing in an infant, but it is **weaker than it was written**, and we say so rather than keep a convenient reason. Whether the 1–736 protein would even function is unresolved (our estimate: ~70/30 that it behaves like the rescuing 731X rather than the failing 753X, from a two-point calibration on orphaned hydrophobic surface). **Best ceiling, no safe way to reach it today.**

So allele 2 is not chosen because it is better. Its ceiling is lower. It is chosen because it is the only one where mechanism and an acceptable safety profile coincide.

3.2 The premise, and how we tested it

Thesis B stands or falls on one claim: **N1002K is a folding variant**, not an interface variant. If it is not, chaperone amplification is aimed at the wrong mechanism and the whole arm collapses.

We tested it by first discovering that the question had been asked wrong. Pathogenic *BUB1B* missense alleles split into two mechanism classes, and stability prediction applies to only one:

class	example alleles	ΔΔG AUC	what is broken
binding interface — TPR, GLEBS	R36Q, T40M, K400E	**0.364** — worse than chance	the partner contact (KNL1, BUB3)
folded core — pseudokinase	R814H, L844F, I909T, L1012P	**0.952**	the fold itself

Pooling them destroys the signal (whole-protein AUC 0.714, *below* burial's 0.760) — which is what every earlier number that looked unstable actually was. Split correctly, this reproduces **Suijkerbuijk 2010's two-class model** from structure alone, where the

original came from westerns.

The serious objection is that ΔΔG predictors read solvent accessibility, so this could be a burial result wearing a costume. Tested directly — each pathogenic variant compared only against benign variants of similar RSA, permutation null shuffling labels *within* matched sets:

matching window	**±5%**	**±10%**	**±15%**	**±20%**
ProteinMPNN-ddG AUC	0.879	0.889	**0.920**	0.917
permutation *p*	0.0052	0.0046	0.0008	**0.0006**

Stable at every window; logistic regression gives ΔΔG a slope of **−2.53** against burial's **−0.22**. Three predictors agree on ranks at Spearman 0.88–0.91, and one (EvoEF2) is a force field rather than a network, so the agreement is not shared training data. **N1002K is placed 2nd, 3rd and 4th worst** of the eleven corpus variants in its domain by the three methods.

What is supported and what is not. *Destabilising* is supported. *Severely destabilising* is not — magnitude is method-dependent. The mechanism is **positional, not chemical**: lysine is only the 10th worst of 19 substitutions at 1002, but the site is the **21st most intolerant of 285**. "This position cannot take substitution" is supported; "lysine's charge clashes" is not. The neighbouring-residue puzzle resolves cleanly — **N1004 ranks 256th of 285**, two adjacent residues at opposite ends of the tolerance distribution.

Confidence that N1002K is a folding variant: **high, but still inference.** Nothing here measures BubR1 abundance in a cell.

And "does it fold?" is the wrong binary. The pharmacological-chaperone field targets mutants that are *"intrinsically active but unstable"*, whose activity is low *"because they are cleared by the quality control system"* — a protein that cannot fold at all is not a candidate; one that folds but spends too long unfolded is exactly the candidate. Destabilising-but-not-severe is the amenable profile, not the hopeless one.

The field's exclusion criterion is that a chaperone is "of no use if the active site is affected", and we measured it (`dock_audit.py`):

residue	**distance to ATP-site centre**
N1002 (proband)	**26.7 Å**
W978 (top VUS nomination)	28.5 Å
L1012 (known pathogenic)	18.7 Å
K795 (VAIK lysine)	2.4 Å

N1002's closest approach to the catalytic lysine is 28.6 Å: **the nucleotide pocket is untouched.** We had been filing that as a disappointment — nothing there to fix. It is the opposite. A lesion in the fold with the site intact is the one configuration this drug class is built for. The same rule **excludes allele 1 outright** — chaperones are "not useful … in which the protein is absent because the gene is affected by … a stop gain mutation" — which is the dosage conclusion of §3.1 reached independently, by the field's own eligibility rule.

3.3 The candidate, and why the ranking inside the class reversed

The axis is **raising folding capacity in *trans***, and two mechanisms do it:

1. **Chemical chaperone** — assists folding directly, which is the actual lesion.

2. **HSP70 / HSF1 induction** — raises quality-control capacity. Treatments that induce Hsp70 restored activity to **17 of 18 (94%)** disease-causing missense CBS mutants, and the same treatments rescued mutant p53 and MTHFR (Singh et al. 2010, PLoS Genet, PMID 20066033). An earlier study rescued four CBS mutants with osmolyte chemical chaperones, and found the rescuable ones were those predicted to *decrease* solvent-accessible surface area (Singh et al. 2007, PMID 17540596).

   The caveat this precedent carries, added 23 Oct 2026 (C47). The three treatments achieving that 94% were **ethanol, proteasome inhibitors, or deletion of the *Hsp26* small heat-shock gene** — and the paper states Hsp70 induction is **"necessary but not sufficient"** for rescue. None of the three is pharmacological HSF1 co-induction: two are not givable to a child and one is a yeast deletion. So the 94% establishes that *this mutation class is rescuable by shifting the chaperone environment*, which is the load-bearing claim. It does **not** establish that raising Hsp70 alone achieves it. Any magnitude transferred from this paper to arimoclomol inherits the insufficiency clause, and the report previously cited the number without it.

Both candidates reach that axis. **4-PBA** does (1) and (2); **arimoclomol** does (2) by co-inducing HSF1, which is the transcription factor aneuploid cells are *depleted of* (§3.3 below, and the reason this class is disease-specific here rather than borrowed).

A third mechanism was originally counted and has been retracted. 4-PBA is also an HDAC inhibitor, and we read that as favourable: HDAC2/3 remove the protective K250 acetylation that shields BubR1 from APC/C degradation, so inhibiting them should *preserve* it. That reasoning is incomplete to the point of being backwards. HDAC inhibition **induces premature sister chromatid separation and overrides the spindle assembly checkpoint**, with loss of MAD2 from the kinetochore, transcription-independently (Magnaghi-Jaulin 2007, Cancer Res 67:6360–67, doi:10.1158/0008-5472.CAN-06-3012), and abrogates BubR1 kinetochore localisation and phosphorylation specifically (Dowling 2005). This child's syndrome is named for premature chromatid separation. 4-PBA is a weak, millimolar HDAC inhibitor, so whether the activity is live at therapeutic exposure is genuinely open — but both answers are bad for the argument: live, and it partially phenocopies the disease; not live, and the third leg does not exist. The general rule, which this report had not written down until the audit forced it: **any chronic agent that degrades SAC function is contraindicated in a SAC-deficient child.** Applied honestly it convicted our own lead (`CORRECTIONS-2026-10-06.md`, C5).

Arimoclomol carries the matched mechanism without that liability, and is FDA-approved — Miplyffa, 20 Sep 2024, Niemann–Pick type C, children aged two and over, in combination with miglustat. Four of our own documents had described it as "phase 3 in ALS" with a "thin paediatric history", which was eighteen months out of date (C7). It is not an HDAC inhibitor.

What 4-PBA still has, and why it is kept rather than dropped. Two things arimoclomol does not. First, **selectivity evidence**: in RPE65 missense disease, PBA left wild-type expression unchanged while significantly raising the three mutants — it raises the unstable allele without over-raising normal protein, which matters when the protein is an oncogene-adjacent mitotic regulator. Second, **paediatric depth**: approved for chronic

management of urea cycle disorders, a neonatal-onset indication, decades of use, multiple formulations. That is not a theoretical profile; it is babies, for years — and it reaches below arimoclomol's age-two floor.

So the honest statement is not that one agent won. It is that the class is right, the mechanism-matched member of it has no checkpoint liability and is approved, and the member with the deeper paediatric record carries a hazard pointed directly at this child's syndrome. **Both are hypotheses for follow-up. Neither has ever been tested on BubR1** (§5). **Carried, not buried:** the FDA review of Ravicti (glycerol phenylbutyrate) flagged *"potential neurotoxicity in pediatric patients under two years of age"* and required post-marketing PK studies in infants. Sodium load and palatability are real issues in chronic paediatric dosing.

	mechanism match	magnitude precedent	paediatric safety	verdict
Arimoclomol / HSF1 co-inducers	correct (chaperone amplification), no checkpoint liability	rescues glucocerebrosidase, destabilised rhodopsin	**FDA-approved (Miplyffa, 20 Sep 2024) for NPC in children ≥2**; not an HDAC inhibitor	**lead** (C7)
4-PBA	folding + QC, **but HDAC inhibition phenocopies premature chromatid separation**	94% rescue of CBS missense (PMID 20066033) — but by ethanol/proteasome/*Hsp26*-deletion, with Hsp70 induction "necessary but not sufficient" (C47); mutant-selective; wild-type-sparing	approved, neonatal indication — reaches below age 2	**demoted to stated alternative** (C5)
SIRT2 activation / NAD+	**wrong degradation route**, plus a K250 hazard	strong — but in a hypomorph, not a folding mutant	not used in this indication	**demoted**
Proteasome inhibition	correct route	rescues in vitro	not viable chronically in a child	no
HSP90 inhibitors	**actively harmful**	—	—	**standing refusal**

The SIRT2 demotion is the argument that decides this ranking, and it demotes where this workstream was heading. Suijkerbuijk showed misfolded BubR1 is cleared by **chaperone-dependent quality control** — levels collapse under HSP90 inhibition, proteasome inhibition rescues. The SIRT2/K668 axis is a different route: CBP acetylation driving ubiquitination of the *normally folded* protein, i.e. **physiological turnover**. Our problem is quality-control triage. Not the same pathway. *(A sequence check en route: the literature disagrees on whether the SIRT2 site is K668 or K688. Against `ENSP00000287598`, residue 668 is K and 668–675 is `KLSPIIED`, an exact LxxIxE PP2A-*

*B56 motif; residue 688 is **S**. The "K688" papers have the number wrong.)* SIRT2 also deacetylates the protective K250, and the acetylation-defective knock-in mouse is **embryonic lethal when homozygous** — 339 live animals monitored, no homozygous newborns. Not a residue to perturb in a child whose disease is already chromosomal instability plus cancer predisposition.

HSP90 inhibitors remain refused outright: they remove virtually all destabilised BubR1 — this child's entire supply.

The argument that makes this arm specific to MVA rather than borrowed

Everything above is a mechanism-*class* argument: 4-PBA works in other proteins, so it might work here. One finding makes it specific to this disease.

Aneuploid human cells have consistently lower HSF1 protein, downregulated HSP90, and a profound defect in mounting the heat-shock response — and constitutively active HSF1 rescues their protein-folding defect.

That closes a loop. Destabilised BubR1 → weakened checkpoint → aneuploidy → **HSF1 and HSP90 fall** → less folding capacity → more BubR1 cleared. This child's cells are chronically aneuploid by definition of the diagnosis, so the chaperone capacity that would buffer a destabilised protein **is itself depleted by the phenotype that protein causes.**

This re-founds the HSP70-inducing arm on a stronger claim. Not "raises quality-control capacity", which is true of any cell, but **restores capacity this disease measurably removes**. It also confirms the HSP90 refusal from an unrelated direction: aneuploid cells are already *significantly more sensitive* to HSP90 inhibition than diploids.

What it does not settle, and we are not choosing a side. Depleted capacity could mean a capacity-restoring drug delivers more than the generic 1.5–3× range — or it could be exactly why such a drug underdelivers here. Both readings are available. It is a reason to measure HSF1 and HSP90 in patient cells, not a reason to claim a larger effect. Full treatment in `BINDING-AND-STABILISING.md`.

3.4 What we tried, and failed, on the structural side

We attempted to make this a **structure-guided** proposal rather than a mechanism-class analogy. Two sampling methods agree the BubR1 pseudokinase ATP site carries **cryptic druggable volume**; if a ligand could occupy it, the chaperone idea becomes a pose. Ten ligands were docked blind into the domain — ATP as positive control, metformin as negative, deucravacitinib because it binds the TYK2 **pseudokinase** domain and is the nearest clinical precedent.

Zero of ten landed within 12 Å of the ATP site. ATP itself finished 38 Å away.

A positive control that misses by 38 Å does not refute a target; it says the instrument is wrong. Two artifacts explain it, and both are diagnosable (`DOCKING-AUDIT.md`, reproduce with `dock_audit.py`):

- **The truncation scar.** Five structurally unrelated drugs converged on residues **824–835** — every residue of which is occluded in full-length BubR1 by the 737–764 linker, **209 deleted contacts** across the site. It is not a binding site; it is the scar left by carving the domain out, and it captured the entire top of the score distribution

including the best pose (imatinib, −8.80). On surface that survives intact, nothing scores better than −7.24.
- **A rigid receptor cannot enter a cryptic pocket.** The pocket's **apo volume is 0 Å³** — cryptic means closed in the apo coordinates. Vina holds the receptor rigid. The experiment could not have succeeded.

The run is uninformative about the site, not evidence against it. We report it because reporting that −8.80 pose as a hit — which is what this would have looked like without the occlusion check — would have been a fabricated lead in a document about a real child.

What it changes upstream is the point: nothing. The arm that survives does not need a pocket. 4-PBA does not occupy a site on BubR1; it raises folding capacity in *trans* and never touches the target. That is why §3.3 ranks it on mechanism match and paediatric safety rather than on affinity, and why a failed docking campaign leaves it exactly where it was.

We then ran the protocol that does it properly, and it returned a clean negative. Millet and co-workers rescued uroporphyrinogen III synthase by docking an **allosteric surface site**, fragment-screening it, validating by thermal shift and NMR, and then searching approved drugs for lookalikes — which found **ciclopirox**, an approved antifungal. Run on BubR1 with a validated instrument (`MILLET-PROTOCOL.md`):

- **Site selection excluded every good pocket for cause.** The one open, persistent, cross-method-reproducible pocket is lined by **His880 and Phe912** — the catalytic spine. The other druggable ones are lined by **Leu1031/Leu1035**, the CENP-E interface. An inhibitor is not a chaperone. A second, independent detector (LIGSITE on the static structure) found exactly one pocket free of both: **134 Å³ at residues 814–820 / 864–875 / 891–905**, intact in full-length, built from four distant sequence segments and so genuinely native-state selective.
- **The instrument was validated first**: re-docking crystal complexes recovered **3 of 4 poses under 2 Å** (imatinib in ABL at 1.03 Å), and seed-to-seed SD is **0.012 kcal/mol**.
- **757 fragments × 5 boxes. The target site is the best box for zero of them** — identical to a flat-surface control, against 48% for the ATP site. Its median score is 0.15 kcal/mol from bare protein. **0 of 757 clear a 1% empirical false-positive rate** against a null built from the decoys themselves.

The conclusion is stronger than "we found nothing": the BubR1 pseudokinase domain has **no ligandable allosteric site that is not already a functional site**. The site we found is geometrically real but shallow — real ≠ druggable. And the controls behaved exactly as the earlier audit predicted, with the truncation scar scoring near the top, which is what licenses reading this as a measurement rather than another void run.

How far that generalises, stated precisely (added 14 Oct 2026, C26b). All five boxes lie inside residues 766–1050, so read strictly this is a result about one domain, and earlier drafts of this report used it as though it closed the structure-guided route for the whole protein. It does close it, but the argument has to be made rather than assumed: **the pseudokinase domain is the only part of BubR1 that structure prediction models with any confidence.** The degradation-determining lysines — K250, where acetylation shields BubR1 from APC/C, and K668 — sit at pLDDT 47.5 and 30.2, in the long intrinsically disordered middle region that makes BubR1 a scaffold. There is no surface there to place a pocket on. So the negative extends to the whole protein not because we

screened the rest but because the rest is not screenable, and that distinction is the honest form of the claim (`ALLELE2-ROUTES-REEXAMINED.md`, `millet/degron_dockability.py`).

This also closes a route the field has used on another gene with exactly our problem. In Friedreich ataxia — where residual frataxin level likewise sets disease severity and no fold-stabilising pocket exists — **ubiquitin-competing molecules** that bind the substrate and block its own ubiquitination raise frataxin in patient cells. The published recipe uses consensus pocket detection (LIGSITE among eight detectors) and AutoDock/Vina, both of which we have validated, with site selection concentrated on the ubiquitination lysine. Such a molecule need not stabilise anything, so our fragment negative does not exclude it — the disorder at K250 and K668 does.

A route that needs no stabilising site at all. A DUBTAC is a PROTAC run backwards: a ligand for the target linked to **EN523, which binds the deubiquitinase OTUB1**, so the chimera strips the K48 degradation chain instead of adding one. `NJH-2-057` (lumacaftor + EN523) raises ΔF508-CFTR **above what lumacaftor achieves alone**. The requirement drops from *"find a molecule that stabilises the fold"* — which our structure work could not deliver — to *"find a molecule that binds"*. A follow-on platform used a **DNA motif** as the targeting arm, so it need not even be drug-like; BubR1 has four characterised interaction motifs that could serve. **No DUBTAC is approved anywhere and none has been made against BubR1** — this is a research direction, not a proposal for this child. It is here because it is the only route that survives the absence of a druggable pocket.

3.5 What Thesis B does not establish

- **Nobody has ever tested 4-PBA, or any chaperone inducer, on BubR1.** Not once. Every precedent above is a different protein. This is a mechanism-class argument.

- **The magnitude is at the edge.** Chemical chaperones typically deliver 1.5–3× rescue. We need ~2× for the attenuated phenotype and ~3× for carrier-equivalent. That is the optimistic end of the observed range, not the middle.

- **It rests on N1002K being an abundance lesion — quantified, not asserted. P(abundance) = 0.79, 95% CI 0.44–0.98** on the measured kinase-domain panel (5 of 6 destabilising; `scripts/mva/abundance_posterior.py`). Dropping the sole exception gives 0.92 (0.62–1.00), and there is a documented reason to: Q921H sat on the same allele as L844F and the authors attributed causality to L844F, so it was never tested as the only variant on an allele. **The weaker figure is the one that ships**, because discarding the one inconvenient data point is not how a headline number should be obtained. **One physicochemical axis survives external validation; the other does not, and the difference was measured rather than argued.** N1002K is the **only variant in the comparison that buries a NEW formal positive charge**, at RSA 12.9% and pLDDT 91.1; every charged panel member *loses* charge instead. That axis is now externally supported: against **measured** steady-state protein abundance in three independent multiplexed assays (PTEN and TPMT VAMP-seq, NUDT15), burying a charge costs abundance and doing it at the surface costs nothing — ρ −0.210/−0.209/−0.140 at buried positions against +0.037/−0.003/+0.031 at exposed ones, on 4,907 buried measured variants (`burial_dependence_test.py`, CORRECTIONS C41).

  §3.5a — what the lysine specifically breaks. "Buries a charge" is a category, and a category cannot be falsified in a lab, so the model's own atoms were asked which interaction a lysine at 1002 destroys (`n1002k_mechanism.py`, C42).

It is **not** a like-charge repulsion variant. Within 8 Å of the N1002 side chain there is exactly one charged group — Arg999 at 6.95 Å, carrying +0.33 of a guanidinium — for a signed burden of **+0.047 e/Å**, which is negligible, and there are **zero** negative charges, so a buried lysine would have no counter-charge either. The mechanism is desolvation of an *unpaired* buried charge.

What it does break is one bond. N1002's amide makes three hydrogen bonds: `OD1 ← Trp978:N` at 2.83 Å, `ND2 → Val998:O` at 2.83 Å, `ND2 → Trp978:O` at 2.95 Å. Asn carries **both** a donor and an acceptor; lysine's NZ **donates only**. The two ND2 donations could in principle be satisfied by a lysine; the **OD1 ← Trp978:N contact cannot be**, leaving a backbone amide without its acceptor at RSA 13.7%.

This is not the W978 packing claim VUS-NOMINATION.md withdrew. That argued outward from N1002 to say `W978L` was an independent prediction, and died because the contact shell is statistically empty — 5 of its 10 residues above their burial-matched median, 5 below. This runs inward, names one backbone hydrogen bond, and does not depend on the shell.

Tested against all 19 substitutions at position 1002 — groups fixed from the mechanism before the scores were read — AlphaMissense orders them as the mechanism predicts (donor-only-and-positive 0.904, no-polar 0.885, acceptor-and-negative 0.841, **acceptor-and-neutral 0.685**), and `N1002S` at 0.348 is the only substitution there it does not call likely-pathogenic. Three things cut against it and are stated rather than smoothed: 18 of 19 are likely-pathogenic, so this explains gradation and not intolerance; the three worst are Trp, Phe and Pro, meaning the dominant signal at 1002 is **sterics**, the axis that failed external validation; and `N1002Q` scores 0.907 while keeping both donor and acceptor, so preserving an acceptor is not sufficient. Ordering *within* one position is also a weaker claim than ranking across the protein, which is the claim C43 shows AlphaMissense cannot support here.

N1002K's side-chain volume change of **+54.5 Å³** was previously offered as a second, independent axis. **It is withdrawn as support.** On the same measured-abundance data, volume change shows no burial dependence in PTEN or TPMT and *reverses* in NUDT15 (−0.176, CI excluding zero). The number is still true and still inside the panel's destabilising range; what it no longer has is external evidence that it predicts abundance where a residue is buried. One validated axis is a weaker claim than two, and it is the claim we have. **It remains the single load-bearing assumption in Track 2 and it has not been measured in this child.** A posterior over six comparators bounds a belief; it does not observe a protein level, and the CI's lower bound is 0.44.

And the support is narrower than it was, because we tested it (C43). The buried-charge axis is validated against measured protein abundance in three other proteins, but asked whether it transfers to *BUB1B's own* labelled variants it **fails**: 2 of 19 pathogenic missense bury a new charge against 2 of 100 benign — a fivefold odds ratio in the right direction, at p = 0.1193. That failure is structural rather than accidental, since 55 of this gene's 69 pathogenic alleles truncate and only 19 missense carry a pathogenic label at all. So the axis is **not demonstrated in this gene**, and the allele-2 abundance argument rests on cross-protein transfer plus the specific hydrogen bond in §3.5a. Across the whole arm the axis applies to **50 of 1,494** missense variants; for the other 1,444 `missense_arm.tsv` records why it does not apply rather than substituting the volume axis that failed in C41a.

- **4-PBA's HDAC inhibition is broad**, and broad epigenetic modulation in a cancer-predisposed infant raises questions this analysis does not answer.

- **The renal phenotype argues against the lead's formulation, and we did not see it until late.** The record shows **nephrocalcinosis from birth (HP:0000121)**. Sodium phenylbutyrate carries a substantial sodium load, which `STABILISATION-STRATEGY.md` raised only as a palatability and dosing nuisance. Against a kidney already depositing calcium it is a different objection, and it is one the glycerol formulation (Ravicti) was designed to avoid — except that the same FDA review flagged potential neurotoxicity under age two. **Neither formulation is clean here**, and that trade needs a nephrologist rather than this analysis.

- **No molecule proposed here binds BubR1, and none can be.** The fragment screen (§3.4) found no ligandable allosteric site, so this arm rests entirely on *trans*-acting proteostasis and would not be rescued by a better docking run. Any future claim that a small molecule stabilises BubR1 **directly** contradicts this result and needs to answer it.

- **Allele 2 is made as protein — now established by elimination, not assumed.** It could have failed to produce protein three ways: NMD (excluded twice over — no stop codon is created, and the variant is in the final exon where a PTC escapes decay by construction), cryptic splicing (**excluded by measurement 14 Oct 2026**: Pangolin gain +0.01 / loss 0.00, stable at window widths of 50, 300 and 1000 nt, with a positive control reproducing its committed score exactly — §3.6, `SPLICE-ALLELE2.md`), or not at all. So the open question is **two-way, not three-way**: abundance or function.

3.6 Where computation stops, and what that means for this submission

This is worth stating directly rather than leaving as an apology in a limitations list, because it defines what the deliverable actually is.

Computation settled more here than is usual, and all of it by exclusion:

question	settled by	result
which allele can a drug reach?	readthrough arithmetic	allele 1 is ~35× short — *quantitatively* insufficient, not merely unproven
is allele 1's transcript destroyed?	calibrated decay model	P(escape) 0.316 — predicted decay, with the residual stated
could allele 2 escape translation?	Pangolin, control-validated	no — gain +0.01, three window widths
is there a druggable site on the target?	757-fragment screen, validated instrument	no — 0 of 757 at 1% FPR
which mechanism class is allele 2 in?	measured comparator panel	folding class at an **83% base rate** (5 of 6)
is any drug class contraindicated by the genotype?	mechanism-derived gate rule	yes — chaperone inhibitors, invisible to any interaction check (§4, C25c)

What computation cannot settle, and no amount more of it would: whether allele 2's lesion is *abundance* (protein scarce, chaperone-amplification on-target) or *function* (protein present, each molecule working poorly, so raising the amount achieves nothing). Those need opposite interventions.

But "cannot settle" is not "has nothing to say", and the distinction matters to anyone deciding whether this arm is worth pursuing. The question is **quantified**: **P(abundance) = 0.79, 95% CI 0.44–0.98**, from the measured kinase-domain panel, with two independent physicochemical axes agreeing (§3.5). Note what does *not* carry the argument: burial. The panel's one tolerated variant is *more* buried than allele 2, so a burial-based reading would get this wrong — what separates them is **buried charge introduction**, the one axis that survives testing against measured protein abundance (C41). Substitution volume was previously named alongside it and is withdrawn: on the same measured data it shows no burial dependence, and reverses in one of three assays. So the honest statement is not "we don't know", and it is no longer "two independent axes agree" either. It is "about four-to-one on, on six comparators, with one externally validated physicochemical axis rather than two, and the lower bound of that interval is 0.44."

So the deliverable is a ranked, falsifiable hypothesis with its discriminating experiment specified — not a drug recommendation. The experiment is cheap and needs no patient material: ectopic LAP-BubR1 N1002K versus wild-type versus **Q921H as the internal negative control**, which is the assay built in 2010 and is a plasmid, a transfection and a blot. It would convert the single largest inference in this workstream into a measurement, and it would also settle the magnitude question that C6 retracted.

And the experiment has a name, a threshold and a regulatory precedent — it is not our invention. Fabry disease has had a formal, FDA- and EMA-sanctioned version of exactly this question for a decade, called **amenability**: a variant is amenable to chaperone therapy if it yields *"abnormally folded or less stable proteins with retained enzymatic activity"*, which is word-for-word the abundance-versus-function distinction. It is decided by a **GLP-validated in vitro assay** — HEK-293 cells transfected with the specific mutation, incubated with drug — and a variant is amenable on a **≥1.2-fold increase over baseline plus a ≥3% absolute increase in wild-type activity**. **600** mutant forms were expressed in HEK-293 and **268** found amenable (Benjamin et al. 2017, Genet Med 19:430-438), and the current product information carries **376** distinct amenable entries, counted off the label by `fabry_precedent.py`. (A figure of "1,386 variants" appeared in earlier drafts of this report, was never sourced, and is retracted — C44.) So our proposed blot should be specified as an amenability determination against those criteria, not as a bespoke experiment.

The same precedent carries an uncomfortable implication we should state rather than leave for a reviewer to notice: **the company that commercialised this strategy built an assay, not a predictor**, and ran it per variant, hundreds of times. That is the field's own verdict on whether computation settles this question, and it agrees with the paragraph above rather than with our posterior. See `HOW-OTHERS-SOLVED-IT.md`.

Our problem is also not unusual, which is the strongest thing to say about generalisability. Reduced protein abundance is *"the most frequent mechanism by which rare missense variants cause disease"*, with surveys putting **40–60% of pathogenic variants** in that class. And in the one protein where the question has been answered exhaustively — the vasopressin 2 receptor — a multiplexed assay over *all possible* variants found more than half of pathogenic ones poorly expressed, and a single stabiliser

rescued 87% of the destabilised set, with the non-rescued variants falling on the drug's own binding site. A single molecule covering most of a variant spectrum is the regime an n-of-1 patient needs.

3.7 The scalable version of this child's question

Everything above answers one question for one variant. The generalisation is a **deep mutational scan of *BUB1B* with an abundance readout, run ± a chaperone co-inducer** — which would classify every *BUB1B* variant as amenable or not **and** test the drug, in one pooled experiment, on the template the V2R study established. That converts an n-of-1 finding into something the next family can use, and it is the honest form of the scalability claim: a proposal with a named precedent, not a result.

One design constraint, which the precedents hide and we should not. Every success in this class — V2R, CFTR, α-Gal A — is a membrane or secreted protein with a trafficking checkpoint, so surface expression or lysosomal delivery gives a readout you can sort cells on. **BubR1 is a nuclear kinetochore scaffold; there is nothing to stain on the surface.** A BubR1 scan needs a tagged-abundance reporter or a mitotic-phenotype readout instead, and choosing between them is real experimental design rather than a detail.

Claiming that computation identified the mechanism would be the error. Declining to name the experiment that would, having narrowed it to one two-way question, would be the other.

The gate gained a genotype-conditional refusal

Added 14 Oct 2026 (C25c). Every other rule in Gate 3 is a fact about the *drug*: withdrawn, carcinogenic, cytotoxic, topical. This one is a fact about the **patient**.

Allele 1 is transcript-null, so this child's entire BubR1 supply is a destabilised missense allele held up by chaperone-dependent quality control — and Suijkerbuijk 2010 measured that HSP90 inhibition removes virtually all destabilised BubR1. An HSP90 inhibitor would therefore deepen the precise deficit causing the disease. **No drug–drug interaction check can see this**, because the interaction is not with another drug; it is with the genotype.

It is implemented as a genotype condition rather than a drug blocklist, so it generalises: the input is "the patient's functional allele is a destabilised missense protein", not "the gene is *BUB1B*". For a patient with two truncating alleles there is no destabilised protein to strip and the rule does not fire. Evidence tiers stay separate — *measured* for HSP90, *inferred* for HSP70/HSF1, where the direction is certain and nobody has measured it on BubR1 — and activators of the same targets are untouched, since co-inducing HSF1 is the therapeutic hypothesis.

It refuses nothing in the current funnel: 13 of 34 survive, unchanged. No HSP90 inhibitor is approved, so Gate 1's approval tiering removes the class before Gate 3 sees it. That is reported rather than left silent, and `tests/test_gate3_chaperone.py` demonstrates in seven cases that the rule can fire — including that it must *not* fire on an activator, which would otherwise refuse our own lead.

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4. The adjunct arm — IL-1 blockade

This was our original lead, and the work behind it stands: a six-gate screen of 5,202 molecules in which three instruments that were not tuned to agree converged on one axis. It is now ranked **second**, for a reason worth stating plainly rather than burying: **it treats a downstream consequence, and Thesis B treats the lesion.** Its premise — that SASP output is elevated in this child — rests on two human datasets that disagree (§5), and unlike Thesis B's premise we had no way to test it here. It is retained, not demoted to a footnote, because if the folding thesis is wrong this is what is left, and because the safety gate below is the most transferable part of the whole method. *(Within §4 this arm is called **Arm A**, the funnel's own terminology, alongside the cancer-predisposition and ciliopathy arms it was run against.)*

4.1 Method — a six-gate funnel

Each gate asks one question, is implemented as a committed script, and reports what it excluded as well as what it passed.

Gate	Question	In	Out
0	Is it a real, publishable molecule?	5,202	4,945
1	Is it an **approved medicine**?	4,945	1,485
2	Does it act on a target an MVA mechanism runs through?	1,485	34
3	Is it safe **in this child**?	34	13
4	Does the thesis survive adversarial attack?	13	13 (1 branch refuted)
5	Has it ever been given to a child?	13	11 studied, **2 leads**

Gate 1 replaced a filter we had planned and found wrong: requiring a WHO ATC code is not an approval test, because ATC classifies substances irrespective of approval. Approval is now tiered by regulator evidence, and the excluded tier is retained so exclusions stay auditable.

Gate 2 is run in **both directions** — target → drug and drug → mechanism — because Open Targets' clinical-candidate list is curated rather than exhaustive. This caught a systematic blind spot: rapalogs are annotated on **FKBP1A, not MTOR** (sirolimus binds FKBP12, and the complex inhibits mTORC1), so a target set naming MTOR misses sirolimus and everolimus entirely. Any drug whose annotated target is a proximal binding partner rather than the effector is invisible to single-direction target screening.

4.2 Gate 3 — the safety gate refused our best candidates

This is the centre of the analysis, not a compliance step.

Arm A's strongest mechanistic hits and the worst safety risks are **the same molecules**. Every TNF blocker (adalimumab, infliximab, etanercept, golimumab) and most JAK inhibitors (tofacitinib, baricitinib, upadacitinib, abrocitinib) carry a black-box **carcinogenicity** warning. In a child whose tumour risk arises from chromosome missegregation — where immunosurveillance is part of what contains it — that is disqualifying rather than a caution. Seven further candidates were refused as ATC L01 antineoplastics **on mechanism**: adding DNA damage to cells that already missegregate chromosomes is the wrong direction, independent of toxicity grade.

We deliberately carried the mitotic targets (AURKA, PLK1, TTK/MPS1, KIF11, WEE1) into Gate 2 so the safety gate could be seen refusing them. **All return zero approved drugs** — every agent against them is investigational. No approved-drug screen can propose a mitotic poison to this child. The danger was real but unreachable.

A finding that inverts the access argument. None of the 13 survivors has a biosimilar, because all five biosimilar-available molecules were refused on carcinogenicity. For this child the affordable options are unsafe and the safer options are expensive orphan biologics. **Access is the binding constraint on the answer, not a closing remark.**

4.3 Gate 4 — we tried to kill our own thesis

The strongest objection is not immunosuppression. It is mechanistic:

> Senescence is a tumour-suppressive barrier. A child who continuously missegregates chromosomes depends on it. Suppressing senescence biology could remove a defence and *increase* tumour risk.

The objection has real support — Li et al., *Oncotarget* 2016 (DOI) conclude that p53-mediated arrest, senescence and apoptosis "control genomic stability in vivo."

The thesis survives because the objection conflates two separable things. Senescence *arrest* is the protective half. The **SASP** is the paracrine output of cells that have already arrested, and it is tumour-promoting in surrounding tissue: Coppé et al., *PLoS Biology* 2008 (DOI) show SASP from **genotoxic-stress-induced** senescence — the state chronically aneuploid cells occupy — drives invasiveness "by a paracrine mechanism that depended largely on the SASP factors interleukin (IL)-6 and IL-8."

An anti-IL-1 or anti-IL-6 antibody does not stop a damaged cell arresting. It blunts what the arrested cell secretes. **The barrier stays; the tumour-promoting signal falls.**

What this refutes

Intervention	Effect on the barrier	Verdict
SASP blockade (anti-IL-1, anti-IL-6)	arrest intact	survives
Senolysis (navitoclax, dasatinib+quercetin, BCL-2 inhibition)	**barrier removed**	**contested — see below**

We ran this a second time, and it refuted us

Round 1 called senolysis a *positive refutation* and this report led with it. A second adversarial round, searching specifically for what had been done in the BubR1 model rather than reasoning from senescence biology in general, found the opposite.

Baker et al., *Nature* 2011 (DOI) — the foundational senescent-cell-clearance paper — was performed **in the BubR1 progeroid mouse**, the closest available model to this child's genotype: "life-long removal of p16(Ink4a)-expressing cells delayed onset of these phenotypes… late-life clearance attenuated progression of already established age-related disorders." Clearing senescent cells in a BubR1-deficient animal **helped**.

The claim is **downgraded to contested, not reversed**, because three things limit Baker as a counter: cancer incidence was not its primary endpoint, and tumour risk was our actual concern; progeroid phenotypes are not the embryonal-tumour axis that dominates this child's risk; and INK-ATTAC is genetic ablation, not a drug with navitoclax's or dasatinib's toxicity in a child.

Honest statement: in the closest animal model senolysis improved healthspan; whether it raises or lowers tumour risk in a SAC-deficient child is unresolved, and our original claim was overstated. Full record in `GATE4-ROUND2-SELF-REFUTATION.md`.

The same search strengthened Arm A's premise

Rao, Asch & Yamada, *Molecular Carcinogenesis* 2016 (DOI) states the mechanism this report had inferred: mitotically damaged cells activate p21/p27/p16 and drive "tissue-level senescence/aging through inflammatory paracrine mechanisms called SASP", and with weakened immune surveillance under systemic CIN, surviving senescent and aneuploid cells produce "an inflammatory, pro-tumorigenic, and senescent/aging microenvironment."

That cuts both ways, and we say so: it supports SASP blockade *and* it supports senolysis. The two are better described as untested alternatives aimed at the same pathology from different ends than as opposites.

The remaining objection, tested

Does blunting inflammation raise cancer risk? For IL-1 specifically the largest randomised test went the other way. CANTOS (Ridker et al., *Lancet* 2017, n=10,061, DOI) reports dose-dependent reductions in incident lung cancer (HR 0.61 at 150 mg, **HR 0.33 at 300 mg**) and lung cancer mortality (HR 0.23 at 300 mg).

We do not claim CANTOS transfers. Different population, cancer and mechanism of risk. It defeats only the general claim that IL-1β blockade is cancer-permissive. And its cost travels with the claim: in the same trial "fatal infections or sepsis were significantly more common in the canakinumab groups," and all-cause mortality did not differ (HR 0.94, 95% CI 0.83–1.06). **An IL-1 proposal trades infection risk for a hypothesised oncological benefit. That trade is the proposal.**

4.4 What the funnel returned

Gate 5 counted paediatric exposure from ClinicalTrials.gov eligibility ages rather than asserting it. 11 of 13 survivors have been studied in children at phase 2+.

	anakinra	canakinumab
Mechanism	IL-1 receptor antagonist	anti-IL-1β monoclonal
Paediatric studies (phase 2+)	**29**	**28**
Youngest eligibility seen	neonatal	neonatal
Gate 3 verdict	caution_immunosuppression	caution_immunosuppression
Carcinogenicity black box	none	none

The convergence was not engineered. Gate 3 kept the IL-1 axis from regulatory drug-warning data; Gate 4 found it is the one mechanism whose large randomised evidence points away from cancer risk; Gate 5 found the same two molecules are the best characterised in children. Three instruments, three data sources, one answer.

A risk this arm did not price in. Gate 3 flagged both leads `caution_immunosuppression` on their labels. The clinical record adds a specific reason: this child has been **treated for rhabdomyosarcoma**, so prior cytotoxic therapy and its effect on marrow reserve and infection risk compound an IL-1 blocker's own signal — and CANTOS already showed fatal infections and sepsis significantly more common under canakinumab. The trade §4.3 describes gets worse, not better, in a treated oncology patient. Any n-of-1 protocol has to be written against the actual treatment history, which this analysis does not have.

Proposed follow-up, in order — and note that step (1) is the *same experiment* that would settle Thesis B, on the same cells, which is the practical argument for running both arms off one sample: (1) measure IL-1β/IL-6 and hsCRP **in primary patient cells, not immortalised lines**, and include an interferon/NF-κB readout — the cell-type caveat above is why, and the MAD1L1 single-cell data is why those extra readouts belong in the panel. This is the missing evidence and it is measurable; (2) if elevated, an n-of-1 protocol with explicit infection monitoring; (3) tumour surveillance continues regardless, since nothing proposed here treats the underlying checkpoint defect.

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5. Limitations

On Thesis B (the lead):

- **No chaperone inducer has ever been tested on BubR1.** Every precedent is a different protein. This is a mechanism-class argument, and it is the single largest gap in the lead.
- **The magnitude ask sits at the edge of the precedent.** ~2× for the attenuated phenotype, ~3× for carrier-equivalent, against a typical chemical-chaperone rescue of 1.5–3×.
- **Everything rests on a predicted structure.** No experimental structure covers N1002 at the resolution ΔΔG methods assume, and nothing here measures BubR1

abundance in a cell — it predicts folding thermodynamics, one contributor among several.
- **Five pathogenic anchors carry the folding-class calibration.** They are source-confounded (4 of 6 positives asserted by UniProt alone, 0 of 28 negatives), mitigated only because four carry direct functional measurements rather than annotations. The negatives are annotations, not measurements.
- **There is no ligandable allosteric site, and this is now measured rather than assumed.** Our first structural attempt was *void* — a blind, rigid dock that scored its own truncation scar and never entered the ATP site (apo volume 0 Å³). We then ran the Millet protocol properly on a validated instrument, and it returned a **clean negative**: the one groove free of both the catalytic spine and the CENP-E interface is the best box for **0 of 757 fragments**, indistinguishable from bare protein, and **0 of 757 clear a 1% empirical false-positive rate** (§3.4, `MILLET-PROTOCOL.md`).
- **So the *pharmacological* chaperone version of this arm is closed**, and the distinction now matters: a pharmacological chaperone binds the target directly and needs a pocket we do not have. Both candidates are **proteostasis regulators acting in *trans*** — 4-PBA chemically, arimoclomol through HSF1 — and neither binds BubR1, which is why the lead survives a negative that kills the structure-guided route. The report should not be read as proposing anything that occupies a site on this protein.
- **The cryptic pocket's geometry is not reproducible between sampling methods.** They agree the ATP site carries cryptic druggable volume; they disagree on where and how much. Quote the claim, not the numbers — and note the screen found even the *open* pockets unremarkable, so cryptic volume is not a latent opportunity here.
- **The allele-1 route is closed on today's drugs, not permanently.** A selective NMD inhibitor with paediatric safety data would immediately outrank everything here on its +40-unit ceiling.

On Thesis A (the adjunct):

- **The tissue channel is observed but DISCORDANT, and this is what demoted this arm.** An earlier draft said no MVA expression cohort exists; that was wrong. GSE22206 profiles a homozygous *BUB1B* patient with confirmed MVA against controls and heterozygous relatives. Scoring Phoenix's axes on it puts the **Immune (SASP) axis DOWN in the proband (d = −1.75), dose-ordered across the allele series** — the opposite of Arm A's prediction. Three things bound that: n=2 per arm makes Cohen's d descriptive only; the cells are EBV-immortalised lymphoblastoid lines, which perturbs the very programmes being measured; and the better human dataset points the other way (below). Full analysis in `PHOENIX-HEADTOHEAD.md`.
- **Counterweight, on better tissue.** Villarroya-Beltri et al., *Science Advances* 2022 (DOI) profiled a biallelic *MAD1L1* SAC-loss patient — an MVA variant — by single-cell RNA on **primary blood cells**, reporting "systemic inflammation with enhanced interferon and NFκB signaling both in aneuploid and euploid cells, suggesting a non-cell autonomous response." That is a paracrine mechanism in the direction Arm A predicts, and NF-κB is the regulator through which IL-1β signals.
- **The two human datasets disagree, and the disagreement is explicable by cell type** — immortalised lines versus primary cells. That is a real limit of the evidence, not a tie to split. The verdict stays `interesting_under_supported`: it does not rise on MAD1L1, and one n=2 immortalised-line comparison cannot sink it.
- **Trial channel remains empty.** No trial of any lead in a chromosomal instability syndrome.
- Paediatric trial eligibility is not approval, not safety in this child, and not evidence in a chromosomal instability syndrome.
- Gate 1's excluded tier contains false negatives — celecoxib is approved but absent from the centralised EMA table we screened on.

- Drug-name resolution reached 989 of 1,485 (67%); misses concentrate in vaccines, immunoglobulins and combination products.
- Metformin and dasatinib are geroprotection/senolytic candidates our target-first method structurally cannot reach.

Cross-cutting:

- **Both alleles are now confirmed in the proband's own callset** (this replaces the earlier limitation that they were not). `proband/confirm_variants.py` queries the challenge VCF directly: both sites are present, **heterozygous, FILTER=PASS, at maximum genotype quality**, at high read depth with allele balances consistent with germline heterozygosity rather than mosaicism, which matters in a mosaic-aneuploidy syndrome. The script prints only these two loci and writes no extract: the DTA forbids resharing, so what is committed is the query, not the data.

- **Trans configuration is still not proven, and this is now the precise gap.** The two sites are **10,911 bp apart** and the callset carries no physical phasing for either (GATK `PGT`/`PID` are null; short-read fragments cannot span that distance). Compound heterozygosity therefore remains an inference — from the diagnosis, and from the fact that a *cis* arrangement would leave one intact allele and would not cause disease. **The clinical record adds independent support**: parental recurrent spontaneous abortion (HP:0200067) is what a heritable biallelic mechanism produces, with each parent carrying one allele — and is not what *cis* predicts. Still not proof. **Parental genotyping remains the single cheapest decisive experiment available**, and it is the one we would run first.

- **The non-coding half of the gene IS scored, and this limitation was false.** Corrected 23 Oct 2026; the earlier text claimed "456 intronic and 612 synonymous ClinVar records… neither thesis currently sees them." Both halves were wrong. The denominators were stale (`clinvar_noncoding.tsv` holds **429** intronic, **563** synonymous and **13** canonical splice-site records, 1,005 in all), and the set was not merely seen but fully scored: `data/nmd/eligibility_map.tsv` carries **1,005 rows, every one with a direct Pangolin call plus exon frame arithmetic and none pending**, yielding **287 survivors** (60 `call`, 227 `candidate`) of which **40** fall in the `candidate_hypomorph_protein_made` tier — the exact class Thesis B addresses. The four known MVA-causing splice alleles are in it as scored calls (`c.751+1G>T`, gain 0.87 / loss −0.90; `c.2009+1G>A`, 0.73 / −0.89).

  What the real limitation is, and it is a sensitivity limitation. The arm is externally calibrated against MFASS on a sealed split (`data/nmd/mfass_benchmark_metrics.json`, n_scored **4,946**): **sensitivity 0.357** (95% CI 0.327–0.388), **specificity 0.986** (0.982–0.989), precision 0.860. So the map is a high-precision, low-recall screen: a positive call is usually right, and it misses roughly two thirds of real splice disruption. A negative is close to uninformative. That is the honest limitation and it should have been this bullet all along.

  Why this correction matters more than the others. A limitation asserting that an analysis does not exist is worse than an unstated one: it tells a reader the work was not done. This arm is the only component of the workstream with stated operating characteristics measured on held-out data in a different gene set, which makes it the most transferable thing here — and the report was actively denying it.

- **Source quality was a first-order problem, not a housekeeping one.** The protein-level variant aggregate carries seven distinct defect classes, all found by audit: six records typed `missense` whose allele is a stop codon, 41 whose wild-type residue

contradicts the reference sequence (nucleotide letters in an amino-acid field — invisible because A, C, G and T are all valid residue codes, and one reached a nomination list), and 84 filed with cDNA coordinates in the protein-position field. All are handled once, in `variants/bub1b.py`. **No single source is complete and none fail the same way.**

Six defects of our own were found and fixed during this work — a target-annotation blind spot, an ATC-group-as-route error, a trial age-unit error, a regex that silently truncated frameshift variants, a residue check that rejected a real insertion allele, and a variant that passed every automated filter while belonging to a different gene. **Each was caught only by cross-checking one instrument against another**, which is an argument for building funnels that disagree with themselves — and the docking audit in §3.4 is the same principle applied to a result we wanted to be true.

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6. Scalability

The funnel (§4) is a parameterised script over public APIs (Open Targets, Ensembl VEP, ClinicalTrials.gov). Changing the mechanism arms — a target list with a written rationale per target — retargets it at another gene or disease without touching the machinery. The paediatric and safety gates are disease-agnostic. It applies wherever a rare-disease genotype implies a pathway and one needs to know which approved drugs reach it, and which must not.

The stability axis (§3) scales differently, and further. Once calibrated on a gene's own pathogenic and benign alleles, it ranks every other variant in the same domain for free. Applied here it nominates **30 of 203 pseudokinase-domain missense VUS at a 4% empirical false-positive rate**, thresholded by specificity against 25 benign controls rather than against the weakest pathogenic anchor. ClinVar holds 1,497 missense VUS in *BUB1B* alone. The nominations are hypothesis-generating and folding-mechanism only — and the top ones follow exactly N1002K's pattern, ordinary substitutions at extraordinarily intolerant positions (`W978L`: 8th worst of 19 at a position ranked 3rd of 285; `N1002K`: 10th worst of 19 at a position ranked 21st).

The transferable method is the two-class split, not the predictor. The result that makes this work is that ΔΔG separates pathogenic from benign at AUC 0.952 inside the folding class and 0.364 — worse than chance — across binding interfaces, so pooling them destroys the signal. Any gene with both a folded domain and a protein-interaction surface has this problem, and most variant-effect pipelines pool. Splitting first, then testing against a **burial-matched** null so the result cannot be solvent accessibility in disguise, is the part worth copying.

And the negative controls are part of the method, not an afterthought. Four results in this report exist only because a control was run: the N1002K pocket effect that benign controls reproduced exactly (structure preparation, not biology); the packing shell that burial-matching showed was unconstrained; the docking campaign whose best-scoring pose sat on our own truncation scar; and the fragment screen, where decoy boxes turned "our site's best fragment scores −5.64" into "that fragment has a 34% empirical false-positive rate" (§3.4).

The transferable part is the shape of that last one. Dock the target *and* the protein's other real pockets, rank the difference rather than the score, and build the null from the decoys themselves. It is the same move as the burial-matched permutation above — control for the confounder, then ask whether anything survives — and it is what separates

a ligandability measurement from a ranked list of greasy molecules. Validating the instrument first (crystal re-docking, seed variance) is the other half: a systematically mistyped receptor is wrong at every box equally, so no decoy could reveal it.

Methods disclosure

Provider: Anthropic. **Interface:** Claude Code (cloud sessions). **Models:** Claude Opus 5, with a short final segment on Claude Opus 5.5. **Setting:** commercial terms, no training on customer content. **Plan/tier:** **`[TEAM TO COMPLETE BEFORE SUBMISSION]`** — the organisers require provider, plan-or-tier and setting; the tier is an account fact this analysis cannot verify about itself.

The AI's role was substantive and is stated plainly: it wrote every script, performed the literature synthesis, and drafted these documents. It also produced the errors listed under Limitations and, in several cases, the controls that caught them. All numbers derive from committed scripts over public APIs, re-derived from raw records by `verify_report.py`; all literature claims cite primary sources.

Full answers to the organisers' methods-description form are in **`METHODS-DESCRIPTION.md`** (and `methods_description_track2_filled.xlsx`).

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